

OKOLAI OTONBARA ALFRED

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SKILLS

CAD: AutoCAD, SketchUp, Tekla TEDS

GIS: QGIS, Google Earth Engine

Hydraulic Modelling: EPA SWMM

Programming Language: Python

Productivity: Microsoft Office (Excel, Word, PowerPoint)

EDUCATION

Bells University of Technology

Feb 2021 – July 2025

BEng in Civil Engineering

EXPERIENCE

Eretek Engineering Nigeria Ltd. – Yenagoa, Nigeria

July 2023 – Dec 2023

Civil Engineering Intern

- Supported the supervision of a 4km stormwater drainage and 2km road network for NDPHC, ensuring strict adherence to design specifications.
- Reviewed technical construction drawings and checked site progress to meet quality and safety benchmarks.
- Collaborated with senior engineers and contractor teams on material planning and performance monitoring.

Koe Procurement and Services Ltd. – Yenagoa, Nigeria

Aug-2022 – Jun 2023

Civil Engineering Intern

- Contributed to the successful delivery of a 40-unit estate development project (5-bedroom duplexes).
- Assisted with site layout, reinforcement placement, and structural supervision for one-storey residential buildings.
- Maintained project timelines through effective coordination with labour teams and site contractors.

PROJECTS

Detailed technical documentation, interactive GIS dashboards, and full project reports are available at otonbara-eng.vercel.app.

Flood Susceptibility Mapping for Bayelsa State (Machine Learning)

- Developed an end-to-end machine learning pipeline on Google Earth Engine to map flood vulnerability using Sentinel-1 SAR and CHIRPS rainfall data.
- Trained a Random Forest classifier (500 trees) achieving a 98% recall and 0.9971 AUC-ROC on a balanced dataset of 70,026 training samples.
- Identified elevation and distance to river as primary predictors (38.5% total model relevance) using MDI and SHAP importance analysis.
- Deployed an interactive React dashboard on Vercel to host static GeoJSON susceptibility maps and model diagnostics.

LID Retrofit Strategy for Opolo Housing Estate, Bayelsa State

- Leveraged QGIS and EPA SWMM to model the hydrologic response of a 15.23-hectare urban catchment under a 10-year design storm.
- Simulated a sustainable retrofit using permeable pavements and rain barrels, achieving an 85% reduction in total runoff and 87% reduction in peak flow.
- Increased catchment retention capacity by over 40x through decentralized source-control LID implementation.
- Delivered comprehensive hydrologic-hydraulic comparisons between baseline 'grey' systems and LID-integrated SuDS.

PUBLICATIONS

- Olanipekun, A. A., Idusuyi, D. U., Alao, F., Bayode, O., Aiyelokun, O. O., Okolai, O. A., & Braimah, J. I. (2025). Analysis and Design of Sustainable Storm-water Management Practice in Ota, Ogun State, Nigeria. *Journal of Renewable Energy and Environment*, 12(2), 153-162.